

Practical Machine Learning and Deep Learning

D1.1 Progress Submission

Team "Magic Photo"

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1 Project Topic

This project objective is to enhance existing solutions for image restoration from different corruptions and coloring of grayscale images using machine learning and deep learning techniques. The project will focus on developing a user-friendly application that allows users to upload images, select the processing option they want to apply, and receive the processed image as output. Additionally, we plan to collect users' feedback for future product improvements and keep track of software lifecycle. The application will leverage pre-trained models and fine-tune them.

Target users are everyone because we want to develop a user-friendly application. The value is that it will allow to restore and colorize not only images, but also important and warm moments of life that were captured in photos.

We assume that for coloring people will use mostly faces, cities, pets, and nature images. For restoration this assumption is not so obvious and strict, the exact focus is under discussion now.

2 Link to GitHub repository

The link to the GitHub repository: <https://github.com/ivaniinno/Image-Restoration-and-Coloring>

3 Existing Solutions

We discovered several promising solutions:

- <https://www.kaggle.com/code/mdzaidanwar/simple-gfpgan-implementation> - A simple implementation of Generative Facial Prior Generative Adversarial Network (GFPGAN) for image restoration.
- <https://www.img2go.com/colorize-image> - This site can be used as a comparison benchmark for image coloring.
- <https://picfix.ai/> - More advanced solution for image processing with a bunch of features.
- <https://vanceai.com/image-enhancer/> - Another site for increasing picture resolution and quality.
- <https://www.myheritage.com/photo-enhancer> - One more solution, which is focused on faces and family photos.

4 Dataset

Since we focus on people, we would like to use the following dataset with photo links and captions: <https://huggingface.co/datasets/nickpai/coco2017-colorization/viewer/default/train?row=18&views%5B%5D=train> (and a version with photos only: <https://huggingface.co/datasets/ayushshah/coco-2017-image-colorization/viewer/default/train?row=18&views%5B%5D=train>)

Also, a sample (for performance reason) of **celebfaces** is a good option.

5 Project Success Criteria

We want to compare the results of models right out of the box and after fine-tuning from our side. The good result would be to achieve stable and solid relative to the original (out of the box) improvement in accuracy, Structural similarity index measure (SSIM), Peak signal-to-noise ratio (PSNR), and to have at least 6/10 positive feedbacks from users on average.

6 Work Distribution

- Maxim Ilin: finding and processing datasets
- Rail Sabirov: MLOps, API, and Frontend
- Ivan Ilyichev: GitHub maintenance, reports, slides
- Everyone: finding and fine-tuning appropriate models with proper logging

7 Project Sharing among Courses

For now we do not plan to share this project with other courses.